CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 14: Equation of a Line

Every formula and result of the chapter on one sheet

1	Slope from the inclination	$m = \tan\theta$
2	Slope through two points	$m = \dfrac{y_2 - y_1}{x_2 - x_1}$
3	Parallel lines	$m_1 = m_2$
4	Perpendicular lines	$m_1 \times m_2 = -1$
5	Collinear points	Slope of AB = slope of BC
6	Slope-intercept form	$y = mx + c$
7	Point-slope form	$y - y_1 = m(x - x_1)$
8	Two-point form	$y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
9	General form $ax + by + c = 0$	Slope $= -\dfrac{a}{b}$; y-intercept $= -\dfrac{c}{b}$
10	x-intercept	Put $y = 0$
11	y-intercept	Put $x = 0$
12	Lines parallel to the axes	$x = a$ (parallel to the y-axis); $y = b$ (parallel to the x-axis)
13	Perpendicular bisector of AB	Passes through the mid-point of AB with slope $-\dfrac{1}{\text{slope of } AB}$